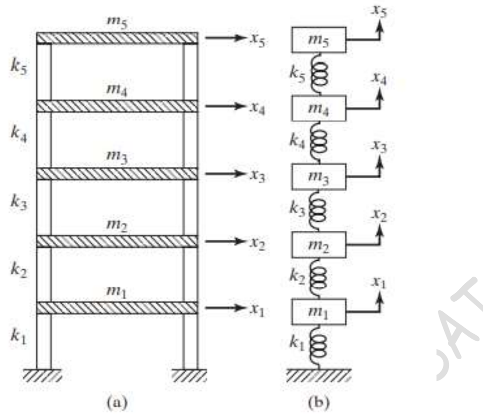
Mass Elements

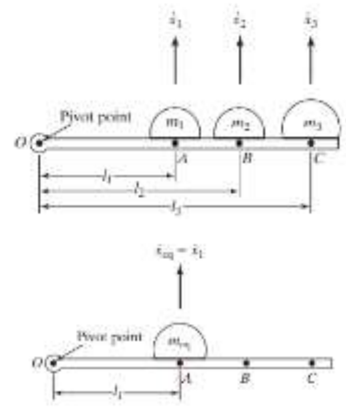
The mass or inertia element is assumed to be rigid body which can gain or lose kinetic energy when its velocity changes.

Consider the example of a multistory building. The vibration (due to wind load or earth quake) will be in the horizontal direction. The columns are the spring elements. Neglecting the mass of the columns compared with mass of the floor, the multi-story building can be modelled as a multidegree freedom system as shown.



Equivalent mass

Consider that three masses are attached to a rigid bar that is pivoted at one end as shown. Let us assume that the three masses are replaced by an equivalent mass m_{eq} placed any where on the rod. For example, let the equivalent mass be at A. Let the velocities of the masses be $\dot{x}_1, \dot{x}_2, and\ \dot{x}_3$



If we look at the translational motion, the kinetic energy of the system,

$$T_{sys} = \frac{1}{2}m_1\dot{x}_1{}^2 + \frac{1}{2}m_2\dot{x}_2{}^2 + \frac{1}{2}m_3\dot{x}_3{}^2$$

The kinetic energy of the equivalent system,

$$T_{eq} = \frac{1}{2}m_{eq}\dot{x}_1{}^2$$

For both systems to be equivalent, the kinetic energy should be the same.

$$T_{eq} = T \ \ggg\gg \ \frac{1}{2}m_{eq}\dot{x}_1{}^2 = \frac{1}{2}m_1\dot{x}_1{}^2 + \frac{1}{2}m_2\dot{x}_2{}^2 + \frac{1}{2}m_3\dot{x}_3{}^2$$

But, we can write

$$\theta = \frac{x_1}{l_1} = \frac{x_2}{l_2} = \frac{x_3}{l_3} \ \ggg\gg \ x_2 = \frac{l_2}{l_1}x_1 \ \ and \ \ x_3 = \frac{l_3}{l_1}x_1$$

$$\dot{x}_2 = \frac{l_2}{l_1}\dot{x}_1 \ \ and \ \ \dot{x}_3 = \frac{l_3}{l_1}\dot{x}_1$$

Hence,

$$m_{eq} = m_1 + m_2\left(\frac{l_2}{l_1}\right)^2 + m_3\left(\frac{l_3}{l_1}\right)^2$$

If we look at the rotational motion, the kinetic energy of the system,

$$T_{sys} = \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}I_2\dot{\theta}^2 + \frac{1}{2}I_3\dot{\theta}^2$$

The kinetic energy of the equivalent system,

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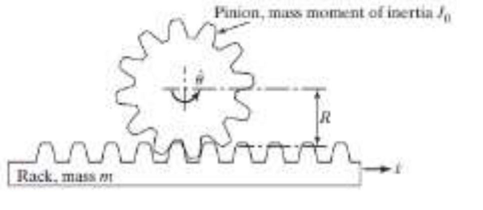
For both systems to be equivalent, the kinetic energy should be the same.

$$T_{eq} = T \ \ggg\gg \ \frac{1}{2}I_{eq}\dot{\theta}^2 = \frac{1}{2}I_1\dot{\theta}^2 + \frac{1}{2}I_2\dot{\theta}^2 + \frac{1}{2}I_3\dot{\theta}^2$$

$$I_{eq} = I_1 + I_2 + I_3 \ \ggg\gg \ m_{eq}l_1{}^2 = m_1l_1{}^2 + m_2l_2{}^2 + m_3l_3{}^2 \ \ggg\gg \ m_{eq} = m_1 + m_2\left(\frac{l_2}{l_1}\right)^2 + m_3\left(\frac{l_3}{l_1}\right)^2$$

Translational and rotational masses

Consider a rack and pinion arrangement as shown with mass and linear velocity of rack m and $\dot{x}$ respectively and mass moment of inertia and angular velocity of pinion as J_0 and $\dot{\theta}$. The pitch circle radius of the pinion is R.



The kinetic energy of the system, $T_{sys} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}J_0\dot{\theta}^2$. We can also see that $x = R\theta$ »» $\dot{x} = R\dot{\theta}$

If we are cnsidering the single equivalent translational mass, $T_{eq} = \frac{1}{2}m_{eq}\dot{x}^2$

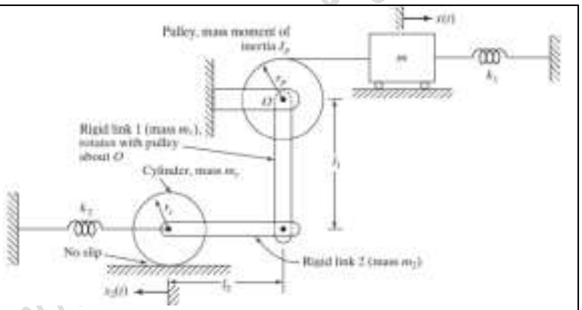
$$T_{eq} = T_{sys} \quad \lll\ggg \quad \frac{1}{2}m_{eq}\dot{x}^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}J_0\dot{\theta}^2 \quad \ll\gg \quad m_{eq} = m + \frac{J_0}{R^2}$$

If we are cnsidering the single equivalent rotational inertia, $T_{eq} = \frac{1}{2}J_{eq}\dot{\theta}^2$

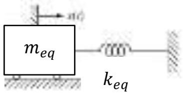
$$T_{eq} = T_{sys} \quad \lll\ggg \quad \frac{1}{2}J_{eq}\dot{\theta}^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}J_0\dot{\theta}^2 \quad \ll\gg \quad J_{eq} = mR^2 + J_0$$

Example 1

Find the equivalent mass and equivalent stiffness of the system shown and repesent it as single degree of freedom system with displacement $x(t)$.



Equivalent system



$T_{eq} = \frac{1}{2}m_{eq}\dot{x}^2 \quad \ll\gg \quad U_{eq} = \frac{1}{2}k_{eq}x^2$

When the mass m has a displacement $x(t)$,

$$\textit{the angle turned by the pulley along with the link 1 will be}, \theta = \frac{x}{r_p}.$$

$$\textit{translation of the cylinder mass centre} = \textit{displ of rigid link 2} = \textit{the displ of the end of the link 1}$$

$$x_2 = l_1\theta = l_1\left(\frac{x}{r_p}\right)$$

$$\textit{rotation of the cylinder}, \theta_{cyl} = \frac{x_2}{r_c} = \left(\frac{l_1}{r_c r_p}\right)x$$

$$\textit{Kinetic energy of the system}, T_{sys} = T_{mass}^{tran} + T_{pulley}^{rot} + T_{link1}^{rot} + T_{link2}^{tran} + T_{cyl}^{tran+rot}$$

$T_{mass}^{tran} = \frac{1}{2}m\dot{x}^2$	$T_{pulley}^{rot} = \frac{1}{2}J_p\dot{\theta}^2$	$T_{link1}^{rot} = \frac{1}{2}I_1\dot{\theta}^2$	$T_{link2}^{tran} = \frac{1}{2}m_2\dot{x}_2^{\,2}$	$T_{cyl}^{tran+rot} = \frac{1}{2}m_c\dot{x}_2^{\,2} + \frac{1}{2}J_c\dot{\theta}_{cyl}^{\,2}$

$$T_{sys} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}J_p\left(\frac{\dot{x}}{r_p}\right)^2 + \frac{1}{2}\left(\frac{m_1l_1^2}{3}\right)\left(\frac{\dot{x}}{r_p}\right)^2 + \frac{1}{2}m_2\left(\frac{l_1\dot{x}}{r_p}\right)^2 + \frac{1}{2}m_c\left(\frac{l_1\dot{x}}{r_p}\right)^2 + \frac{1}{2}\left(\frac{m_cr_c^2}{2}\right)\left(\frac{\dot{x}l_1}{r_cr_p}\right)^2$$

$$\textit{Strain energy of the system}, U_{sys} = U_{sprg1} + U_{sprg2} = \frac{1}{2}k_1x^2 + \frac{1}{2}k_2x_2^{\,2} = \frac{1}{2}k_1x^2 + \frac{1}{2}k_2\left(\frac{l_1x}{r_p}\right)^2$$

For equivalent systems,

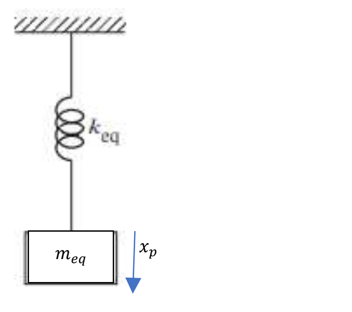
$$\frac{1}{2}m_{eq}\dot{x}^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}J_p\left(\frac{\dot{x}}{r_p}\right)^2 + \frac{1}{2}\left(\frac{m_1l_1^2}{3}\right)\left(\frac{\dot{x}}{r_p}\right)^2 + \frac{1}{2}m_2\left(\frac{l_1\dot{x}}{r_p}\right)^2 + \frac{1}{2}m_c\left(\frac{l_1\dot{x}}{r_p}\right)^2 + \frac{1}{2}\left(\frac{m_cr_c^2}{2}\right)\left(\frac{\dot{x}l_1}{r_cr_p}\right)^2$$

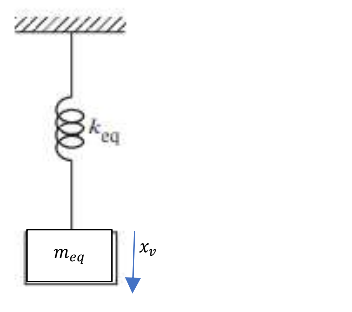
$$\frac{1}{2}k_{eq}x^2 = \frac{1}{2}k_1x^2 + \frac{1}{2}k_2\left(\frac{l_1x}{r_p}\right)^2$$

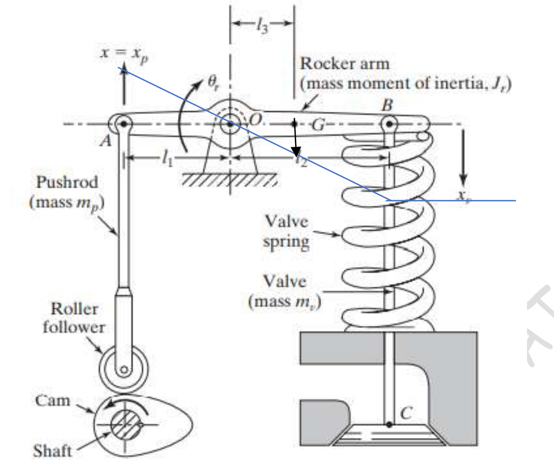
$$m_{eq} = m + \frac{J_p}{r_p^{\,2}} + \frac{m_1l_1^2}{3r_p^{\,2}} + \frac{m_2l_1^2}{r_p^{\,2}} + \frac{m_cl_1^2}{r_p^{\,2}} + \frac{m_cl_1^2}{2r_p^{\,2}} \quad \ll\gg \quad k_{eq} = k_1 + k_2\frac{l_1^2}{r_p^{\,2}}$$

Example 2

In the cam-foloower mechanism, the follower system consists of a push rod of mass m_p, a rocker arm of mass m_r and mass moment of inertia I_G about its centre of gravity, a valve of mass m_v and a valve spring of negligible mass. Find the equivalent mass of the system by assuming its location at A and C.







When the pushrod with mass m_p has a displacement x_p,

$$\text{the angle turned by the rocker arm will be, } \theta_r = \frac{x_p}{l_1}.$$

$$\text{translation of the rocker arm mass centre, } x_g = l_3\theta_r = \left(\frac{l_3}{l_1}\right)x_p$$

$$\text{displacement of the valve, } x_v = l_2\theta_r = \left(\frac{l_2}{l_1}\right)x_p$$

$$\text{Kinetic energy of the system, } T_{sys} = T_{pushrod}^{tran} + T_{rocker}^{tran+rot} + T_{valve}^{tran} +$$

$T_{pushrod}^{tran} = \frac{1}{2}m_p\dot{x_p}^2$	$T_{rocker}^{tran+rot} = \frac{1}{2}m_r\dot{x_g}^2 + \frac{1}{2}I_g\dot{\theta_r}^2$	$T_{valve}^{tran} = \frac{1}{2}m_v\dot{x_v}^2$

If the equivalent mass is placed at A,

$$T_{sys} = \frac{1}{2}m_p\dot{x_p}^2 + \frac{1}{2}m_r\left(\frac{l_3}{l_1}\dot{x_p}\right)^2 + \frac{1}{2}I_g\left(\frac{\dot{x_p}}{l_1}\right)^2 + \frac{1}{2}m_v\left(\frac{l_2}{l_1}\dot{x_p}\right)^2$$

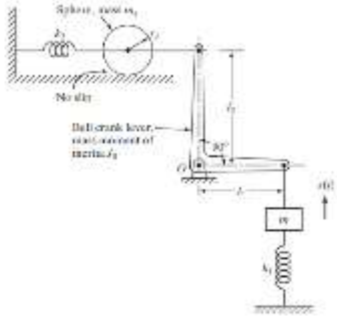
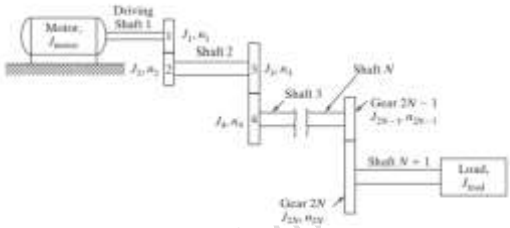
$$\frac{1}{2}m_{eq}\dot{x_p}^2 = \frac{1}{2}m_p\dot{x_p}^2 + \frac{1}{2}m_r\left(\frac{l_3}{l_1}\dot{x_p}\right)^2 + \frac{1}{2}I_g\left(\frac{\dot{x_p}}{l_1}\right)^2 + \frac{1}{2}m_v\left(\frac{l_2}{l_1}\dot{x_p}\right)^2 \quad \text{«»} \quad m_{eq} = m_p + m_r\left(\frac{l_3}{l_1}\right)^2 + \frac{I_g}{l_1^2} + m_v\left(\frac{l_2}{l_1}\right)^2$$

If the equivalent mass is placed at C,

$$T_{sys} = \frac{1}{2}m_p\left(\frac{l_1}{l_2}\dot{x_v}\right)^2 + \frac{1}{2}m_r\left(\frac{l_3}{l_1}\frac{l_1}{l_2}\dot{x_v}\right)^2 + \frac{1}{2}I_g\left(\frac{\frac{l_1}{l_2}\dot{x_v}}{l_1}\right)^2 + \frac{1}{2}m_v\dot{x_v}^2$$

$$\frac{1}{2}m_{eq}\dot{x_v}^2 = \frac{1}{2}m_p\left(\frac{l_1}{l_2}\dot{x_v}\right)^2 + \frac{1}{2}m_r\left(\frac{l_3}{l_2}\dot{x_v}\right)^2 + \frac{1}{2}I_g\left(\frac{\dot{x_v}}{l_2}\right)^2 + \frac{1}{2}m_v\dot{x_v}^2 \quad \text{«»} \quad m_{eq} = m_v + m_r\left(\frac{l_3}{l_2}\right)^2 + \frac{I_g}{l_2^2} + m_p\left(\frac{l_1}{l_2}\right)^2$$

******* Find out the equivalent spring stiffness in both cases ******

Exercise 1 Convert the given system in to single degree of freedom spring-mass system with equivalent mass and equivalent stiffness having output $x(t)$	
Exercise 2 Find the equivalent inertia of the gear train shown with reference to the driving shaft. J_i and n_i represent the mass moment of inertia and number of teeth of gear $i = 1,2, \dots \dots .\ 2N$	

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition